From Data Products to Inference: The Econometrics of Fuzzy Joins

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Abstract

Data products built on fuzzy joins often ship a single canonical linkage while concealing the candidate-match graph from which that linkage was constructed. This note studies downstream regression when the analyst can observe the candidate graph. Two natural single-dataset resolutions fail for different reasons. Edge expansion introduces an algebraic variance penalty that mechanically shrinks the slope relative to a score-collapsed regression. Score averaging, by contrast, induces nonclassical measurement error whose bias need not attenuate and can have either sign. Which linkage errors matter depends on the estimand: joins that assign treatment status, joins that attach outcomes to pre-assigned groups, and level estimands face different error costs. I discuss two practical corrections—multiple imputation over candidate assignments and validation-sample regression calibration—and then show that both may miss a second problem: upstream contamination when the covariates shipped in the target table were themselves created by an earlier fuzzy join. The note's main message is that linkage uncertainty is not a single scalar property of a linked file; it is estimand-specific, and some of it is inherited from the data product itself.

1 Introduction

Empirical researchers increasingly work with linked data products assembled upstream from administrative or transactional files. In many applications the linking step is not exact: records are joined using names, dates, locations, and other quasi-identifiers that produce a set of plausible candidate matches rather than a single certainty. The analyst, however, usually receives a clean final file and proceeds as if each row were linked without error.

This note asks a downstream question. Suppose the data producer ships not only a canonical linkage but the candidate-match graph itself: for each source record, all plausible

target records together with a pairwise score. How should an analyst convert that object into a regression dataset, and what happens if they do so naively?

The existing literature on regression with linked data is substantial. Early bias-correction work includes Scheuren and Winkler (1993, 1997) and Lahiri and Larsen (2005). Subsequent contributions address probability-linked data, incomplete linkage, probabilistic multi-linkage, correlated linkage error, and linked-data secondary analysis more broadly (Chambers, 2009; Kim and Chambers, 2012b,a, 2015; Han and Lahiri, 2019; Chambers and Diniz da Silva, 2020; Wang et al., 2022; Doidge and Harron, 2019). Multiple-imputation and primary-analysis approaches include Goldstein et al. (2012); Harron et al. (2014); Hof and Zwinderman (2012, 2015); Hof et al. (2017); Tancredi and Liseo (2011, 2015); Gutman et al. (2013); Steorts et al. (2016); Dalzell and Reiter (2018). More recent post-linkage work includes Slawski et al. (2025); Slawski et al. (2021) is a closely related precursor for partially shuffled linear regression rather than a general linked-data framework. A companion paper (Sood, 2025) formalizes the join itself as a selection problem over feasible linked tables, connecting to the finite selection model literature; this note takes up the downstream inference question that follows.

My contribution is narrower. I focus on the econometric problem created when a data product ships a many-to-many candidate graph for a fuzzy geographic join. I make four points. First, the inferential consequences of linkage error depend on the estimand. Second, the two most obvious one-shot resolutions of the candidate graph produce distinct pathologies. Third, multiple imputation and validation-based calibration are useful but answer different questions. Fourth, even perfect treatment of the analyst's own candidate graph does not solve upstream linkage error frozen into the shipped covariates.

2 Setup and notation

Let $S = \{1, \ldots, n\}$ index source records and $T = \{1, \ldots, K\}$ index target records. For source unit i, let the true target be $v(i) \in T$. The regression of interest is

$$Y_i = \beta_0 + \beta_1 X_{v(i)} + \eta_{v(i)} + \varepsilon_i, \tag{1}$$

where X_k is a target-level covariate, η_k is a target-level shock, and ε_i is idiosyncratic noise.

A fuzzy matcher produces a candidate graph $\mathcal{G} = (S, T, E, s)$, where $E \subseteq S \times T$ is the set of candidate edges and $s : E \to [0, 1]$ is a pairwise score. For source unit i, define the candidate set

$$\mathcal{C}_i = \{(j, s_{ij}) : (i, j) \in E\}.$$

I distinguish throughout between the raw pairwise score s_{ij} and a calibrated match probability

$$p_{ij} \approx P(v(i) = j \mid \mathcal{C}_i),$$

which is an additional object that must be estimated; it is not implied by the score itself.

A central theme of the paper is the mismatch between the *edge-level* nature of the candidate graph and the *row-level* structure of the regression dataset (see Sood, 2025, for a general treatment of this mismatch). The graph contains many candidate edges per source unit. The regression requires one covariate and one cluster assignment per source unit.

3 Which linkage errors matter depends on the estimand

The first mistake analysts make is to talk about "linkage error" as though it had a universal inferential cost. It does not.

Treatment-assignment joins. If the linkage determines treatment status, a false match contaminates both groups: one unit is moved into the wrong group and another is displaced from the correct one. In that setting, false positives and false negatives both matter directly for bias.

Outcome-attachment joins within pre-assigned groups. If the groups are fixed before linkage and the join only attaches outcomes within group, then within-group permutations are benign for unweighted group means. Only errors that cross an estimand-relevant boundary matter for bias.

Level estimands. If the estimand is a level for a specific population, every wrong attachment matters.

The practical implication is that the "best" resolution of a candidate graph is estimand-specific. A treatment-assignment design typically requires a much harsher trade-off against false matches than a design in which ambiguity remains within groups.

When calibrated probabilities are available, the resolution problem can be written as a maximum-weight matching problem. Let the expected utility of selecting edge (i, j) be

$$a_{ij} = u_{\mathrm{TP}}\, p_{ij} - u_{\mathrm{FP}}(1 - p_{ij}),$$

where $u_{\mathrm{TP}} > 0$ is the value of a true positive and $u_{\mathrm{FP}} > 0$ is the cost of a false positive. Then the linkage rule solves

$$\max_{L \subseteq E} \sum_{(i,j) \in L} a_{ij} \tag{2}$$

subject to the application's assignment constraints. Edges with $a_{ij} \leq 0$ can be screened out ex ante, but the final solution remains a constrained optimization problem rather than a pure edgewise threshold rule.

4 Two naive resolutions of the candidate graph

4.1 Expanded edge-level regression

A natural first idea is to keep every edge (i, j) with score above a threshold and run a weighted regression on the resulting edge-level file. Let

$$w_{ij} = \frac{s_{ij}}{\sum_{j':(i,j') \in E} s_{ij'}} \qquad \text{with} \qquad \sum_j w_{ij} = 1.$$

Define the score-weighted average covariate

$$\tilde{X}_i = \sum_j w_{ij} X_j$$

and the within-candidate-set variance

$$V_i = \sum_j w_{ij} (X_j - \tilde{X}_i)^2.$$

Proposition 1 (Expanded versus collapsed slope)**.** *Assume the regression is fit with centered variables and edge weights w_{ij}. The weighted least-squares slope from the expanded edge-level file is*

$$\hat{\beta}_{\mathrm{exp}} = \frac{\sum_i Y_i \tilde{X}_i}{\sum_i \tilde{X}_i^2 + \sum_i V_i}.$$

If instead one collapses to one row per source unit using covariate $\tilde{X}_i$, the resulting slope is

$$\hat{\beta}_{\mathrm{col}} = \frac{\sum_i Y_i \tilde{X}_i}{\sum_i \tilde{X}_i^2}.$$

Hence, whenever the two slopes share the same sign,

$$|\hat{\beta}_{\text{exp}}| \leq |\hat{\beta}_{\text{col}}|.$$

Proof. Because Y_i is constant across the candidate edges for source unit i,

$$\sum_{i,j} w_{ij} Y_i X_j = \sum_i Y_i \tilde{X}_i.$$

Also,

$$\sum_{i,j} w_{ij} X_j^2 = \sum_i \tilde{X}_i^2 + \sum_i V_i$$

by the usual variance decomposition identity. □

This is an algebraic result, not a subtle asymptotic one. Expanding the graph and reusing the same outcome across multiple candidate covariates mechanically adds the penalty $\sum_i V_i$ to the denominator. The more heterogeneous the candidate set, the larger the penalty.

The expanded file also corrupts the cluster structure. A target cluster in the expanded file contains outcomes generated under other targets' shocks whenever those other units include the target as a candidate. That makes the expanded graph a useful shipped artifact but a poor regression dataset.

4.2 Score-collapsed regression

A second natural idea is to collapse each candidate set to a single averaged covariate,

$$\tilde{X}_i = \sum_j w_{ij} X_j.$$

Write

$$\tilde{X}_i = X_i^* + u_i, \qquad X_i^* \equiv X_{v(i)}.$$

The quantity u_i is measurement error induced by score averaging.

Proposition 2 (Collapsed regression as nonclassical EIV)**.** *Suppose*

$$E[\eta_{v(i)} + \varepsilon_i \mid X_i^*, u_i] = 0.$$

Then the probability limit of the collapsed OLS slope is

$$\operatorname{plim} \hat{\beta}_{\text{col}} = \beta_1 \frac{\operatorname{Var}(X^*) + \operatorname{Cov}(X^*, u)}{\operatorname{Var}(X^*) + \operatorname{Var}(u) + 2\operatorname{Cov}(X^*, u)}.$$

If $\mathrm{Cov}(X^*, u) = 0$*, this reduces to classical attenuation. If* $\mathrm{Cov}(X^*, u) \neq 0$*, the bias can have either sign.*

Proof. Write $Y_i = \beta_1 X_i^* + \nu_i$, where $\nu_i = \eta_{v(i)} + \varepsilon_i$ and $E[\nu_i \mid X_i^*, u_i] = 0$. Then

$$\operatorname{plim} \hat{\beta}_{\mathrm{col}} = \frac{\mathrm{Cov}(X^* + u, \beta_1 X^* + \nu)}{\mathrm{Var}(X^* + u)} = \beta_1 \frac{\mathrm{Var}(X^*) + \mathrm{Cov}(X^*, u)}{\mathrm{Var}(X^*) + \mathrm{Var}(u) + 2\,\mathrm{Cov}(X^*, u)}.$$

□

The intuition is straightforward. In many fuzzy matching problems, errors are not exchangeable across all records. They are local to a confusable neighborhood. A unit from *rampur* is likely to be confused with *rampura* or *ramnagar*, not with an arbitrary distant place. Once ambiguity is structured in that way, u_i will generally covary with X_i^*, and attenuation is no longer the default case.

5 Corrections

5.1 Multiple imputation over candidate assignments

When the analyst has access to the candidate graph and calibrated probabilities, a direct fix is to treat the unresolved linkage as missing data. The required conditions are stronger than they are sometimes presented.

Assumption 1 (Conditions for MI over candidate assignments)**.** *The multiple-imputation strategy requires:*

1. *high recall: for most units, the true target lies in the candidate set;*
2. *calibrated probabilities:* p_{ij} *is a reasonable approximation to* $P(v(i) = j \mid \mathcal{C}_i)$*;*
3. *an acceptable approximation to the dependence structure induced by the graph.*

Under these conditions, one can draw M completed datasets by assigning one target per source unit in each draw, estimate the regression in each completed dataset, and combine results with Rubin's rules (Rubin, 1987). This avoids both the within-candidate-set variance penalty of the expanded estimator and the averaging pathology of the collapsed estimator.

The diagnostic $\lambda = (1 + 1/M)B/T$, where B is the between-imputation variance and T is the total variance, is itself estimand-specific. If all ambiguous candidates fall within the same group or have similar covariate values, different draws produce similar estimates and λ is small regardless of the ambiguity share. If the candidates straddle an estimand-relevant

boundary, λ concentrates exactly where it should. The same candidate graph produces different λ values for different downstream analyses.

The main caveat is dependence. If a connected component contains many source units and a small number of targets, independent draws at the unit level may produce implausible assignments. In those settings, a component-level sampler or a joint posterior over matchings is preferable (Tancredi and Liseo, 2011; Steorts et al., 2016).

5.2 Validation-sample calibration

If a validation sample with verified links is available, the problem changes. Let X_i^π denote the covariate attached by the linked file and let X_i^* denote the verified covariate in the validation sample. In the single-regressor linear case, suppose the calibration model is

$$X_i^* = \alpha + \gamma X_i^\pi + r_i, \qquad E[r_i \mid X_i^\pi] = 0.$$

If the substantive model is linear with one mismeasured regressor, then a regression-calibration correction is

$$\hat{\beta}_{\mathrm{RC}} = \frac{\hat{\beta}_{\mathrm{naive}}}{\hat{\gamma}}.$$

This is a scalar correction for a scalar problem. With multiple regressors, interactions, or nonlinear outcome models, one needs multivariate regression calibration or a fuller joint error model rather than simple division (Carroll et al., 2006).

The validation sample must represent the full estimand population, not just the subset retained by a canonical matching rule. Otherwise it cannot reveal selection induced by missed matches.

6 When does matching uncertainty matter?

Three conditions determine whether matching uncertainty is large relative to sampling uncertainty. First, a non-trivial share of records must be ambiguous. Second, the candidate targets in ambiguous sets must differ meaningfully on X. Third, the ambiguity must cross estimand-relevant boundaries.

The first two conditions are about the candidate graph: how many source units have close candidates, and how different those candidates are on the covariate of interest. The third is about the downstream analysis. If the analyst's estimand is a within-group mean and all ambiguous candidates fall within the same group, matching uncertainty is irrelevant even when the first two conditions hold. If the estimand is a treatment effect and ambigu-

ous candidates straddle the treatment boundary, matching uncertainty is concentrated in the most damaging place. The same candidate graph produces different effective matching uncertainty for different estimands.

If any of the three conditions fails, matching uncertainty is small.

Tables 1–3 show the five estimators, MI diagnostics, and a bootstrap variance decomposition in a medium-scale simulation ($K = 40$ targets, $n = 600$ source units, three confusable name clusters with covariate spread up to 0.50, DGP $\beta_1 = 5$).

Table 1: Estimator comparison, medium-scale simulation ($K = 40$, $n = 600$).

Method	$\hat{\beta}_1$	SE	N (rows)	Clusters
True DGP	5.000	—	—	—
Oracle	6.322	1.233	600	40
Canonical	6.051	1.170	576	40
Expanded	3.655	0.329	1892	598
Collapsed	9.214	1.564	598	40
MI (Rubin)	6.005	1.151	—	—

Table 2: MI diagnostics via Rubin's rules ($M = 200$ draws).

Quantity	Value
Imputations (M)	200
Pooled $\hat{\beta}_1$	6.0054
Within-imputation var ($\bar{U}$)	1.3162
Between-imputation var (B)	0.0078
Total var (T)	1.3240
Total SE	1.1507
λ	0.0060
95% CI	[3.750, 8.261]

Table 3: Bootstrap variance decomposition ($B = 200$ replicates).

Component	SD($\hat{\beta}_1$)	Var($\hat{\beta}_1$)
Sampling only	0.4653	0.2165
Matching only	0.0866	0.0075
Joint	0.4316	0.1863
Sum of components	—	0.2240

The collapsed estimator overshoots the oracle by 46%. The expanded estimator undershoots by 42%. MI recovers the oracle to within 5%. The between-imputation share of

variance is $\lambda = 0.006$, reflecting a design in which only 4% of source units are ambiguous. The bootstrap exercise shows that matching uncertainty is small relative to sampling uncertainty in this design: the variance of $\hat{\beta}_1$ from sampling (holding matches fixed) is 0.22, from matching (holding the sample fixed) is 0.008, and the joint variance is 0.18.

In regimes with 15–35% ambiguous records, λ can exceed 0.10. Whether the analyst needs λ or just a corrected estimate determines which correction path to take. As simulation-based guidance: if $\lambda < 0.05$, matching uncertainty can be noted but safely ignored; if $\lambda > 0.10$, the MI-based interval should be the primary reported interval.

7 Implications for data products

The corrections in Section 5 address the analyst's own linkage uncertainty. Two paths are available: MI over the candidate graph, or regression calibration with a validation sample. Both assume the covariates in the target table are measured without error. In practice, many data products are themselves assembled by fuzzy matching, and the analyst inherits a second layer of matching error that neither MI nor regression calibration can address.

7.1 Upstream contamination

Let X_k^{pub} denote the covariate published for target k. Suppose the data product computes X_k^{pub} by averaging administrative records assigned to k in an earlier join J_1. If a fraction α_k of records assigned to k truly belong elsewhere, the published covariate is

$$X_k^{\text{pub}} \approx (1 - \alpha_k) X_k^* + \alpha_k \bar{X}_{-k}, \tag{3}$$

where X_k^* is the true target-level covariate and $\bar{X}_{-k}$ is the mean among the wrongly assigned records. Equation (3) is exact only when the correctly retained records for target k are representative of the target; otherwise it should be read as a first-order approximation. The contamination error is $u_k^{(1)} = \alpha_k(\bar{X}_{-k} - X_k^*)$.

If the analyst then misassigns source unit i to target $\pi(i)$ in their own join J_2, the total covariate error decomposes as

$$u_i = \underbrace{(X^*_{\pi(i)} - X^*_{v(i)})}_{\text{downstream wrong-target error}} + \underbrace{u^{(1)}_{\pi(i)}}_{\text{upstream contamination}}.$$

The two components are not independent: both error sources are structured by the same geography, and when $\pi(i) \neq v(i)$, the upstream contamination is evaluated at the wrong

target. Applying Proposition 2 to the two-layer error,

$$\operatorname{plim}\hat{\beta} = \beta_1 \cdot \frac{\operatorname{Var}(X^*) + \operatorname{Cov}(X^*, u^{(1)} + u^{(2)})}{\operatorname{Var}(X^*) + \operatorname{Var}(u^{(1)}) + \operatorname{Var}(u^{(2)}) + 2\operatorname{Cov}(u^{(1)}, u^{(2)}) + 2\operatorname{Cov}(X^*, u^{(1)} + u^{(2)})}. \tag{4}$$

The analyst can estimate $\operatorname{Var}(u^{(2)})$ from the candidate graph they observe. The remaining terms are frozen into the data product. Consequently, λ computed from MI over J_2 understates total matching uncertainty.

7.2 Estimand-specific contamination

The upstream contamination affects different published covariates differently, even though the underlying matching errors are the same. The misassignment share α_k is the same for all covariates computed from the same upstream-linked records. But the gap $\bar{X}_{-k} - X_k^*$ depends on how much the covariate varies within the confusable cluster.

Proposition 3 (Estimand-specific contamination)**.** *Let $X^{(a)}$ and $X^{(b)}$ be two target-level covariates computed from the same set of J_1-linked records, with contamination errors $u_k^{(1,a)} = \alpha_k(\bar{X}_{-k}^{(a)} - X_k^{*(a)})$ and $u_k^{(1,b)} = \alpha_k(\bar{X}_{-k}^{(b)} - X_k^{*(b)})$. Then*

$$\frac{\operatorname{Var}(u^{(1,a)})}{\operatorname{Var}(u^{(1,b)})} = \frac{\sum_k \alpha_k^2(\bar{X}_{-k}^{(a)} - X_k^{*(a)})^2}{\sum_k \alpha_k^2(\bar{X}_{-k}^{(b)} - X_k^{*(b)})^2}.$$

This ratio depends entirely on the within-cluster heterogeneity of each covariate and can be arbitrarily large or small.

In a companion simulation ($K = 20$, $n = 150$, multivariate DGP with $\beta_{\text{lit}} = 5$ and $\beta_{\text{pop}} = 1.5$), two analysts using the same data product, same joins, and same source units face a 38-fold difference in total bias: -0.14 for the literacy coefficient versus $+0.004$ for the log-population coefficient. The upstream match rate is identical for both analysts. The upstream contamination share α_k is identical. The damage is not.

7.3 What data products should ship

The upstream contamination result has a concrete implication: summary match-quality statistics—overall match rate, pairwise accuracy—are insufficient diagnostics for downstream analysts. Two analysts with the same match rate face different effective contamination, and neither can assess their own exposure from the published match rate alone.

A useful linked-data product should expose three tiers of artifacts.

The baseline tier is the canonical linkage with match-status flags identifying each record as an exact match, unambiguous fuzzy match, ambiguous candidate set, or no candidate. Analysts who never look deeper still benefit from knowing which records are exact matches, which are fuzzy, and which were dropped.

The second tier is the candidate-match graph for the analyst's own join. This enables MI, regression calibration (when combined with a validation sample), and estimand-specific resolution. Without this, the analyst is locked into the canonical policy's implicit $u_{\mathrm{FP}}/u_{\mathrm{TP}}$ ratio. Because different estimands face different error costs (Section 3), different analysts may need to resolve the same candidate graph at different thresholds; shipping a single canonical linkage forecloses that choice.

The third tier addresses upstream contamination. At minimum, the data producer should document which published covariates were computed from fuzzy-matched records, along with the match rate and the geographic distribution of ambiguous matches. Ideally, the data producer would ship a validation sample or the upstream candidate-match table, enabling the downstream analyst to propagate both layers of uncertainty. Without these, the upstream component of matching uncertainty is invisible and frozen.

The broader lesson is that linkage uncertainty is not fully summarized by an overall match rate. What matters is where ambiguity lies, how it covaries with the variables used in the downstream estimand, and how much error was already embedded upstream in the shipped covariates. Matching uncertainty, when non-negligible, belongs in the same accounting as sampling, clustering, and heteroskedasticity—for the analyst's own join. For the upstream join, they need the data producer's help.

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Appendix

A Schema for fuzzy-joined data products

This appendix proposes a relational schema for shipping data products built on fuzzy joins. The schema covers the two-table case: a source table S linked to a target table T via string similarity on noisy identifiers. Eight tables are defined. The first two are inputs; the remaining six are artifacts the data producer ships alongside the analysis-ready panel.

These six artifacts correspond to the three tiers described in Section 6.3 of the main text. The canonical linkage table is the baseline tier: match-status flags for every source record. The candidate-match table, connected component table, and component-covariate diagnostics table form the second tier: the candidate graph and per-covariate diagnostics that enable MI, regression calibration, and estimand-specific resolution. The validation sample table and covariate provenance table support the third tier: verified links for the analyst's own join and documentation of upstream linkage provenance.

All schemas use SQL-style definitions for precision. Primary keys are marked `PK`; foreign keys reference their parent table. Implementers can adapt these to flat files (CSV/Parquet) by treating the key constraints as documentation rather than enforcement.

A.1 Input tables

A.1.1 Source table

One row per record to be linked. The table includes the raw reported name used for matching, outcome variables, and any covariates measured at the source level.

```sql
CREATE TABLE source (
    source_id        INTEGER PRIMARY KEY,
    reported_name    TEXT NOT NULL,
    -- outcome and source-level covariates
    revenue          FLOAT,
    n_employees      INTEGER,
    year             INTEGER
);
```

A.1.2 Target table

One row per entity in the reference frame. The table includes the canonical name, geographic identifiers, and target-level covariates used in downstream regressions.

```sql
CREATE TABLE target (
    target_id        INTEGER PRIMARY KEY,
    target_name      TEXT NOT NULL,
    -- geographic hierarchy
    district_id      INTEGER,
    state_id         INTEGER,
    -- target-level covariates
    literacy         FLOAT,
    population       INTEGER,
    area_sq_km       FLOAT
);
```

A.2 Tier 1: Canonical linkage

A.2.1 Canonical linkage table

One row per source record. This is the default analysis dataset: the deterministic resolution of the candidate-match graph under a stated policy (τ, δ). Every source record appears exactly once, whether matched or not.

```sql
CREATE TABLE canonical_linkage (
    source_id        INTEGER PRIMARY KEY
                     REFERENCES source(source_id),
    target_id        INTEGER
                     REFERENCES target(target_id),
                     -- NULL if unmatched
    match_score      FLOAT,
                     -- score of the assigned pair; NULL if unmatched
    match_status     ENUM('exact',
                          'unambiguous_fuzzy',
                          'ambiguous_dropped',
                          'below_threshold',
                          'no_candidates')
                     NOT NULL,
    match_method     TEXT,
                     -- e.g., 'levenshtein', 'jaro_winkler_v2'
    score_gap        FLOAT,
                     -- s_{i(1)} - s_{i(2)}; NULL if <= 1 candidate
```

```
    n_candidates     INTEGER NOT NULL DEFAULT 0,
    component_id     INTEGER
                     -- connected component in the candidate graph
);
```

Design notes.

- `match_status` distinguishes the reason a record is or is not linked. `exact` means the match was deterministic (e.g., a shared unique identifier). `unambiguous_fuzzy` means the top candidate passed both τ and δ. The remaining three categories are different failure modes: ambiguous (multiple close candidates), below threshold (best score too low), and no candidates (nothing passed screening).
- `target_id` is `NULL` for unmatched records. This makes the canonical linkage a left join from source to target, not an inner join. Every source record is present, so the analyst can see what was dropped and why.
- `n_candidates` and `score_gap` are summary statistics from the candidate set. They allow the analyst to assess match quality at the row level without loading the full candidate table.

A.3 Tier 2: Candidate graph and diagnostics

A.3.1 Candidate-match table

One row per (source, candidate target) pair above the screening threshold ε. This is the bipartite candidate-match graph $\mathcal{G}$. Every resolution policy discussed in the paper—expanded, collapsed, MI, alternative (τ', δ')—can be reconstructed from this table.

```
CREATE TABLE candidate_matches (
    source_id        INTEGER NOT NULL
                     REFERENCES source(source_id),
    target_id        INTEGER NOT NULL
                     REFERENCES target(target_id),
    match_score      FLOAT NOT NULL
                     CHECK (match_score >= 0 AND match_score <= 1),
    match_probability FLOAT,
                     -- calibrated P(v(i)=j | C_i); NULL if
                     -- uncalibrated. See metadata for whether
                     -- this column is populated.
```

```
    match_rank       INTEGER NOT NULL,
                     -- rank within this source_id's candidate set
                     -- (1 = best score)
    is_default       BOOLEAN NOT NULL,
                     -- TRUE if this pair is the canonical assignment
    component_id     INTEGER,
    PRIMARY KEY (source_id, target_id)
);
```

Design notes.

- The primary key is (source_id, target_id): each pair appears at most once.
- `match_score` is the raw pairwise similarity. `match_probability` is the calibrated posterior probability $p_{ij} \approx P(v(i) = j \mid \mathcal{C}_i)$. These are different objects, as emphasized in Section 2 of the main text. When calibrated probabilities are available, they should be shipped alongside the raw scores. When they are not, the column is `NULL` and the metadata should say so. The MI procedure in Section 5.1 and the decision-theoretic formulation in Section 3 (equation 2) both require p_{ij}, not s_{ij}.
- `is_default` flags which pair, if any, is the canonical assignment. For unmatched source records, no row has `is_default = TRUE`. For matched records, exactly one row does. Reconstructing the canonical linkage is: `SELECT * FROM candidate_matches WHERE is_default = TRUE`.
- Source records with zero candidates have no rows in this table. To recover the full source population, left-join `canonical_linkage` to `candidate_matches`.
- The screening threshold ε should be documented in metadata. It is typically looser than τ, so the candidate table includes pairs that the canonical policy would not assign.

A.3.2 Connected component table

One row per connected component in the candidate-match graph. A connected component is a maximal set of source and target records linked by candidate edges. Singletons have no ambiguity. Dense components with multiple sources and targets are where matching decisions are interdependent.

```
CREATE TABLE components (
    component_id     INTEGER PRIMARY KEY,
```

```
    n_sources        INTEGER NOT NULL,
    n_targets        INTEGER NOT NULL,
    n_edges          INTEGER NOT NULL,
    density          FLOAT,
                     -- n_edges / (n_sources * n_targets)
    mean_score       FLOAT,
    min_gap          FLOAT,
                     -- minimum score gap among source records
    n_ambiguous      INTEGER
                     -- sources with status 'ambiguous_dropped'
);
```

Design notes.

- This table is a diagnostic summary. The analyst can quickly identify components where matching uncertainty concentrates: large `n_ambiguous`, high `density`.
- Covariate-specific diagnostics are in the separate `component_covariate_diagnostics` table below, because within-cluster heterogeneity is covariate-specific (Proposition 3 in the main text).

A.3.3 Component-covariate diagnostics table

One row per (component, covariate) pair. This table implements the insight from Proposition 3 in the main text: the same upstream matching error produces different contamination for different covariates, depending on within-cluster heterogeneity. The analyst can query this table to determine whether their specific covariate is at risk.

```
CREATE TABLE component_covariate_diagnostics (
    component_id     INTEGER NOT NULL
                     REFERENCES components(component_id),
    covariate_name   TEXT NOT NULL,
                     -- matches a column name in the target table
    x_range          FLOAT,
                     -- max(X) - min(X) across targets in component
    x_mean           FLOAT,
    x_sd             FLOAT,
    PRIMARY KEY (component_id, covariate_name)
);
```

Design notes.

- `x_range` measures how much MI draws can vary within this component for this covariate. A large range in a component with many ambiguous records means matching uncertainty matters for this covariate. A small range means it does not, regardless of how ambiguous the component is.
- The analyst can join this table with `components` and sort by `n_ambiguous * x_range` to find where matching uncertainty concentrates for their specific estimand.
- This table should be populated for every target-level covariate that appears in the shipped data product. It is the schema-level implementation of the estimand-specific diagnostic discussed in Section 6.2.

A.4 Tier 3: Validation and upstream provenance

A.4.1 Validation sample table

One row per source record in the verified golden set. The golden set is a random sample from the *full source population*, including records that the canonical policy dropped.

```
CREATE TABLE validation_sample (
    source_id        INTEGER PRIMARY KEY
                     REFERENCES source(source_id),
    true_target_id   INTEGER NOT NULL
                     REFERENCES target(target_id),
    verification     ENUM('unique_id', 'clerical_review',
                          'field_verification')
                     NOT NULL,
    canonical_target_id  INTEGER
                     REFERENCES target(target_id),
                     -- NULL if this record was dropped
    match_status     ENUM('exact',
                          'unambiguous_fuzzy',
                          'ambiguous_dropped',
                          'below_threshold',
                          'no_candidates')
                     NOT NULL
);
```

Design notes.

- The critical design requirement (Section 5.2 of the main text) is that this table must sample from the full source population, not just from matched records. Without dropped records, selection bias from false negatives cannot be assessed.
- `canonical_target_id` alongside `true_target_id` lets the analyst compute the confusion matrix and calibration slope directly.
- The golden set should be stratified to oversample from ambiguous components. If stratified, sampling weights should be documented.

A.4.2 Covariate provenance table

One row per target-level covariate in the shipped data product. This table documents which covariates were themselves produced from upstream fuzzy joins, enabling the analyst to assess exposure to the upstream contamination discussed in Section 6.1 of the main text.

```sql
CREATE TABLE covariate_provenance (
    covariate_name  TEXT PRIMARY KEY,
                    -- matches a column name in the target table
    source_join     TEXT,
                    -- identifier for the upstream join that
                    -- produced this covariate, or 'direct' if
                    -- not derived from a fuzzy join
    join_method     TEXT,
                    -- e.g., 'levenshtein', 'probabilistic_rl'
    upstream_match_rate FLOAT,
                    -- share of upstream records successfully
                    -- matched; NULL if source_join = 'direct'
    upstream_n_ambiguous INTEGER,
                    -- upstream records in ambiguous candidate
                    -- sets; NULL if source_join = 'direct'
    upstream_validation_available BOOLEAN,
                    -- whether a validation sample exists for
                    -- the upstream join
    notes           TEXT
);
```

Design notes.

- The key distinction is between covariates that come from a deterministic source (e.g., census tables linked by administrative codes, where `source_join = 'direct'`) and covariates that were themselves computed from fuzzy-matched records. The analyst cannot assess upstream contamination without knowing which covariates fall into which category.
- When `upstream_validation_available = TRUE`, the data producer should ship the upstream validation sample or document where it can be obtained. This enables regression calibration of the upstream error (Section 5.2).
- Combined with the `component_covariate_diagnostics` table, the analyst can answer the question: "For my specific covariate, was it produced by a fuzzy join, and if so, how much within-cluster heterogeneity exists in the components where my source records concentrate?" This is the concrete implementation of the estimand-specific contamination analysis in Proposition 3.

Example rows.

covariate	source_join	method	match rate	n_ambig	valid?
literacy	ec_to_village_v3	levenshtein	0.92	1840	FALSE
population	census_admin_code	direct	NULL	NULL	—
area_sq_km	census_admin_code	direct	NULL	NULL	—

Literacy was computed from an upstream fuzzy join with 92% match rate and 1,840 ambiguous records. Population and area come from census tables linked by administrative codes and carry no upstream matching uncertainty. The analyst regressing on literacy faces upstream contamination; the analyst regressing on population does not. Neither can tell from the overall match rate alone.

A.5 Reconstruction queries

The following queries reconstruct the resolution policies discussed in Section 4 of the main text.

Canonical linkage (analysis-ready panel).

```sql
SELECT s.source_id, s.revenue, t.literacy
FROM source s
JOIN canonical_linkage cl ON s.source_id = cl.source_id
```

```sql
JOIN target t ON cl.target_id = t.target_id
WHERE cl.match_status IN ('exact', 'unambiguous_fuzzy');
```

Canonical linkage under alternative (τ', δ').

```sql
WITH ranked AS (
    SELECT source_id, target_id, match_score, match_rank,
           match_score - LEAD(match_score) OVER (
               PARTITION BY source_id ORDER BY match_rank
           ) AS gap
    FROM candidate_matches
)
SELECT source_id, target_id, match_score
FROM ranked
WHERE match_rank = 1
  AND match_score >= 0.70   -- tau'
  AND (gap >= 0.15 OR gap IS NULL);  -- delta'
```

Estimand-specific resolution via $u_{\mathrm{TP}}/u_{\mathrm{FP}}$. When calibrated probabilities are populated, the analyst can resolve the candidate graph against a decision-theoretic threshold (equation 2 in the main text). A pair is included only when its expected utility is positive:

```sql
-- Estimand-specific resolution
-- Set u_tp and u_fp to reflect the analyst's error costs
WITH scored AS (
    SELECT source_id, target_id, match_probability,
           :u_tp * match_probability
             - :u_fp * (1 - match_probability) AS expected_util,
           ROW_NUMBER() OVER (
               PARTITION BY source_id
               ORDER BY match_probability DESC
           ) AS prob_rank
    FROM candidate_matches
    WHERE match_probability IS NOT NULL
)
SELECT source_id, target_id, match_probability
FROM scored
WHERE prob_rank = 1
  AND expected_util > 0;
```

```
-- Treatment-assignment: set u_fp >> u_tp (e.g., 10:1)
-- Outcome-attachment:   set u_fp ~ u_tp  (e.g., 2:1)
```

Expanded full join.

```sql
WITH totals AS (
    SELECT source_id, SUM(match_score) AS total_score
    FROM candidate_matches
    WHERE match_score >= 0.50
    GROUP BY source_id
)
SELECT cm.source_id, cm.target_id, s.revenue, t.literacy,
       cm.match_score / tot.total_score AS weight
FROM candidate_matches cm
JOIN source s ON cm.source_id = s.source_id
JOIN target t ON cm.target_id = t.target_id
JOIN totals tot ON cm.source_id = tot.source_id
WHERE cm.match_score >= 0.50;
```

Collapsed (score-weighted average X).

```sql
WITH totals AS (
    SELECT source_id, SUM(match_score) AS total_score
    FROM candidate_matches
    WHERE match_score >= 0.50
    GROUP BY source_id
)
SELECT cm.source_id, s.revenue,
       SUM(cm.match_score / tot.total_score * t.literacy)
           AS literacy_avg
FROM candidate_matches cm
JOIN source s ON cm.source_id = s.source_id
JOIN target t ON cm.target_id = t.target_id
JOIN totals tot ON cm.source_id = tot.source_id
WHERE cm.match_score >= 0.50
GROUP BY cm.source_id, s.revenue;
```

MI draw (single imputation). MI draws require randomness and are better implemented in a scripting language. Pseudocode for one draw:

```
-- Pseudocode: one MI draw
FOR EACH source_id IN canonical_linkage:
  IF match_status IN ('exact', 'unambiguous_fuzzy'):
    drawn_target = target_id  -- deterministic
  ELSE IF n_candidates > 0:
    candidates = SELECT target_id, match_probability
                 FROM candidate_matches
                 WHERE source_id = :source_id
                   AND match_probability IS NOT NULL
    -- draw one target proportional to match_probability
    drawn_target = WEIGHTED_SAMPLE(candidates)
  ELSE:
    SKIP  -- no candidates; exclude from this draw
  INSERT INTO mi_draw_m (source_id, drawn_target)
```

Regression calibration from the golden set.

```
-- Calibration slope: regress true X on assigned X
SELECT v.source_id,
       t_true.literacy AS x_true,
       t_canon.literacy AS x_assigned
FROM validation_sample v
JOIN target t_true ON v.true_target_id = t_true.target_id
JOIN target t_canon ON v.canonical_target_id = t_canon.target_id
WHERE v.canonical_target_id IS NOT NULL;
-- Feed into: x_true ~ x_assigned to get gamma_hat

-- Selection bias diagnostic (requires full-population golden set)
SELECT v.source_id, s.revenue, t_true.literacy AS x_true
FROM validation_sample v
JOIN source s ON v.source_id = s.source_id
JOIN target t_true ON v.true_target_id = t_true.target_id;
-- Compare: revenue ~ x_true (all records)
-- versus:  revenue ~ x_true (WHERE canonical_target IS NOT NULL)
-- Gap = selection bias from dropping ambiguous records
```

Upstream contamination assessment. The analyst can estimate the implied contamination share α_k for each target from the candidate graph. For each target k, α_k is approxi-

mated by the fraction of candidate edges pointing to k from source records whose best match is a *different* target:

```
-- Implied contamination share per target
WITH best_matches AS (
    SELECT source_id,
           target_id AS best_target
    FROM candidate_matches
    WHERE match_rank = 1
),
edge_classification AS (
    SELECT cm.target_id,
           CASE WHEN cm.target_id != bm.best_target
                THEN 1 ELSE 0
           END AS is_spillover
    FROM candidate_matches cm
    JOIN best_matches bm ON cm.source_id = bm.source_id
)
SELECT target_id,
       COUNT(*) AS n_edges,
       SUM(is_spillover) AS n_spillover,
       AVG(is_spillover) AS alpha_hat
FROM edge_classification
GROUP BY target_id
ORDER BY alpha_hat DESC;
-- Targets with high alpha_hat are those where upstream
-- contamination is most likely to affect published covariates.
-- Cross-reference with covariate_provenance to identify
-- which covariates are at risk.
```

A.6 Metadata

The data product should ship a metadata file (JSON or plain text) documenting:

- **Matching algorithm:** name, version, parameters.
- **Screening threshold** ε: the minimum score for a pair to appear in `candidate_matches`.
- **Assignment thresholds** (τ, δ): the score and gap thresholds used to produce the canonical linkage.

- **Score interpretation:** what `match_score` represents (e.g., "1 minus normalized Levenshtein distance," "Fellegi-Sunter log-likelihood ratio"), and whether `match_probability` is populated.
- **Calibration method:** if `match_probability` is populated, how it was estimated (e.g., "isotonic regression on validation sample," "multinomial within candidate set").
- **Validation sample design:** sampling frame (full population or matched only), stratification variables, sampling weights if applicable.
- **Coverage summary:** number of source records by match status, overall match rate, number of connected components, and the share of source records in non-singleton components.